Physical Properties of Carbon Nanotubes

R SAITO
UNIVERSITY OF ELECTRO-COMMINUCATIONS, TOKYO

G DRESSELHAUS & M S DRESSELHAUS
MIT

RUCA-BIBLIOTHEEK
03 01 0136534 1

ICP
Imperial College Press

CHAPTER 2.

Tight Binding Calculation of Molecules and Solids

In carbon materials except for diamond, the π electrons are valence electrons which are relevant for the transport and other solid state properties. A tight binding calculation for the π electrons is simple but provides important insights for understanding the electronic structure of the π energy levels or bands for graphite and graphite-related materials.

2.1 Tight Binding Method for a Crystalline Solid

In this section we explain the tight binding method for a crystalline solid. In the following sections, we show some examples of energy bands for carbon materials discussed in the Chapter 1.

2.1.1 Secular Equation

Because of the translational symmetry of the unit cells in the direction of the lattice vectors, $\vec{a}_i$, $(i = 1, \cdots, 3)$, any wave function of the lattice, Ψ, should satisfy Bloch's theorem

$$T_{\vec{a}_i}\Psi = e^{i\vec{k}\cdot\vec{a}_i}\Psi, \quad (i = 1, \cdots, 3), \tag{2.1}$$

where $T_{\vec{a}_i}$ is a translational operation along the lattice vector $\vec{a}_i$, and $\vec{k}$ is the wave vector[45, 46]. There are many possible functional forms of Ψ which satisfy Eq. (2.1). The most commonly used form for Ψ is a linear combination of plane waves. The reason why plane waves are commonly used is that: (1) the integration of the plane wave wavefunction is easy and can be done analytically, (2) the numerical accuracy only depends on the number of the plane waves used. However, the plane wave method also has limitations: (1) the scale of the

computation is large, and (2) it is difficult to relate the plane wave wavefunction to the atomic orbitals in the solid.

Another functional form which satisfies Eq. (2.1) is based on the $j-$th atomic orbital in the unit cell (or atom). A tight binding, Bloch function $\Phi_j(\vec{k},\vec{r})$ is given by,

$$\Phi_j(\vec{k},\vec{r}) = \frac{1}{\sqrt{N}}\sum_{\vec{R}}^{N} e^{i\vec{k}\cdot\vec{R}}\varphi_j(\vec{r}-\vec{R}), \quad (j=1,\cdots,n). \tag{2.2}$$

Here $\vec{R}$ is the position of the atom and φ_j denotes the atomic wavefunction in state j. The number of atomic wavefunctions in the unit cell is denoted by n, and we have n Bloch functions in the solid for a given $\vec{k}$. To form $\Phi_j(\vec{k},\vec{r})$ in Eq. (2.2), the φ_j's in the N ($\sim 10^{24}$) unit cells are weighted by the phase factor $\exp(i\vec{k}\cdot\vec{R})$ and are then summed over the lattice vectors $\vec{R}$ of the whole crystal. The merits of using atomic orbitals in Bloch functions are as follows: (1) the number of basis functions, n, can be small compared with the number of plane waves, and (2) we can easily derive the formulae for many physical properties using this method.* Hereafter we consider the tight binding functions of Eq. (2.2) to represent the Block functions.

It is clear that Eq. (2.2) satisfies Eq. (2.1) since

$$\begin{aligned}\Phi_j(\vec{k},\vec{r}+\vec{a}) &= \frac{1}{\sqrt{N}}\sum_{\vec{R}}^{N} e^{i\vec{k}\cdot\vec{R}}\varphi_j(\vec{r}+\vec{a}-\vec{R}) \\ &= e^{i\vec{k}\cdot\vec{a}}\frac{1}{\sqrt{N}}\sum_{\vec{R}-\vec{a}}^{N} e^{i\vec{k}\cdot(\vec{R}-\vec{a})}\varphi_j(\vec{r}-(\vec{R}-\vec{a})) \\ &= e^{i\vec{k}\cdot\vec{a}}\Phi_j(\vec{k},\vec{r}),\end{aligned} \tag{2.3}$$

where we use the periodic boundary condition for the $M \equiv N^{-1/3}$ unit vectors in each $\vec{a}_i$ direction,

$$\Phi_j(\vec{k},\vec{r}+M\vec{a}_i) = \Phi_j(\vec{k},\vec{r}) \ \ (i=1,\cdots,3), \tag{2.4}$$

consistent with the boundary condition imposed on the translation vector $T_{M\vec{a}_i} = 1$. From this boundary condition, the phase factor appearing in Eq. (2.2) satisfies $\exp\{ikMa_i\} = 1$, from which the wave number k is related by the integer p,

$$k = \frac{2p\pi}{Ma_i}, \quad (p=0,1,\cdots,M-1), \ \ (i=1,\cdots,3). \tag{2.5}$$

In three dimensions, the wavevector $\vec{k}$ is defined for the x, y and z directions, as k_x, k_y and k_z. Thus $M^3 = N$ wave vectors exist in the first Brillouin zone, where the k_i can be considered as continuum variables.

The eigenfunctions in the solid $\Psi_j(\vec{k},\vec{r})$ $(j=1,\cdots,n)$, where n is the number of Bloch wavefunctions, are expressed by a linear combination of Bloch functions $\Phi_{j'}(\vec{k},\vec{r})$ as follows:

$$\Psi_j(\vec{k},\vec{r}) = \sum_{j'=1}^{n} C_{jj'}(\vec{k})\Phi_{j'}(\vec{k},\vec{r}), \tag{2.6}$$

where $C_{jj'}(\vec{k})$ are coefficients to be determined. Since the functions $\Psi_j(\vec{k},\vec{r})$ should also satisfy Bloch's theorem, the summation in Eq. (2.6) is taken only for the Bloch orbitals $\Phi_{j'}(\vec{k},\vec{r})$ with the same value of $\vec{k}$.

The $j-$th eigenvalue $E_j(\vec{k})$ $(j=1,\cdots,n)$ as a function of $\vec{k}$ is given by

$$E_j(\vec{k}) = \frac{\langle\Psi_j|\mathcal{H}|\Psi_j\rangle}{\langle\Psi_j|\Psi_j\rangle} = \frac{\int\Psi_j^*\mathcal{H}\Psi_j d\mathbf{r}}{\int\Psi_j^*\Psi_j d\mathbf{r}} \tag{2.7}$$

where $\mathcal{H}$ is the Hamiltonian of the solid. Substituting Eq. (2.6) into Eq. (2.7) and making a change of subscripts, we obtain the following equation,

$$E_i(\vec{k}) = \frac{\sum_{j,j'=1}^{n} C_{ij}^* C_{ij'}\langle\Phi_j|\mathcal{H}|\Phi_{j'}\rangle}{\sum_{j,j'=1}^{n} C_{ij}^* C_{ij'}\langle\Phi_j|\Phi_{j'}\rangle} \equiv \frac{\sum_{j,j'=1}^{n}\mathcal{H}_{jj'}(\vec{k})C_{ij}^*C_{ij'}}{\sum_{j,j'=1}^{n}\mathcal{S}_{jj'}(\vec{k})C_{ij}^*C_{ij'}} \tag{2.8}$$

where the integrals over the Bloch orbitals, $\mathcal{H}_{jj'}(\vec{k})$ and $\mathcal{S}_{jj'}(\vec{k})$ are called transfer integral matrices and overlap integral matrices, respectively, which are defined by

$$\mathcal{H}_{jj'}(\vec{k}) = \langle\Phi_j|\mathcal{H}|\Phi_{j'}\rangle, \ \ \mathcal{S}_{jj'}(\vec{k}) = \langle\Phi_j|\Phi_{j'}\rangle \ \ (j,j'=1,\cdots,n). \tag{2.9}$$

When we fix the values of the $n\times n$ matrices $\mathcal{H}_{jj'}(\vec{k})$ and $\mathcal{S}_{jj'}(\vec{k})$ in Eq. (2.9) for a given $\vec{k}$ value, the coefficient C_{ij}^* is optimized so as to minimize $E_i(\vec{k})$.

*The limitations of the tight binding method are that: (1) there is no simple rule to improve the numerical accuracy and (2) atomic orbitals do not describe the interatomic region.

It is noted that the coefficient C_{ij}^* is also a function of $\vec{k}$, and therefore C_{ij}^* is determined for each $\vec{k}$. When we take a partial derivative for C_{ij}^* while fixing the other $C_{ij'}$, $C_{ij'}^*$, and C_{ij} coefficients,[†] we obtain zero for the local minimum condition as follows,

$$\frac{\partial E_i(\vec{k})}{\partial C_{ij}^*} = \frac{\sum_{j'=1}^{N} \mathcal{H}_{jj'}(\vec{k}) C_{ij'}}{\sum_{j,j'=1}^{N} \mathcal{S}_{jj'}(\vec{k}) C_{ij}^* C_{ij'}} - \frac{\sum_{j,j'=1}^{N} \mathcal{H}_{jj'}(\vec{k}) C_{ij}^* C_{ij'}}{\left(\sum_{j,j'=1}^{N} \mathcal{S}_{jj'}(\vec{k}) C_{ij}^* C_{ij'}\right)^2} \sum_{j'=1}^{N} \mathcal{S}_{jj'}(\vec{k}) C_{ij'} = 0. \tag{2.10}$$

When we multiply both sides of Eq. (2.10) by $\sum_{j,j'=1}^{N} \mathcal{S}_{jj'}(\vec{k}) C_{ij}^* C_{ij'}$ and substitute the expression for $E_i(\vec{k})$ of Eq. (2.8) into the second term of Eq. (2.10), we obtain

$$\sum_{j'=1}^{N} \mathcal{H}_{jj'}(\vec{k}) C_{ij'} = E_i(\vec{k}) \sum_{j'=1}^{N} \mathcal{S}_{jj'}(\vec{k}) C_{ij'}. \tag{2.11}$$

Defining a column vector,

$$\mathbf{C}_i = \begin{pmatrix} C_{i1} \\ \vdots \\ C_{iN} \end{pmatrix}, \tag{2.12}$$

Eq. (2.11) is expressed by

$$\mathcal{H}\mathbf{C}_i = E_i(\vec{k})\mathcal{S}\mathbf{C}_i. \tag{2.13}$$

Transposing the right hand side of Eq. (2.13) to the left, we obtain $[\mathcal{H} - E_i(\vec{k})\mathcal{S}]\mathbf{C}_i = 0$. If the inverse of the matrix $[\mathcal{H} - E_i(\vec{k})\mathcal{S}]$ exists, we multiply both sides by $[\mathcal{H} - E_i(\vec{k})\mathcal{S}]^{-1}$ to obtain $\mathbf{C}_i = 0$ (where 0 denotes the null vector), which means that no wavefunction is obtained. Thus the eigenfunction is given only when the inverse matrix does not exist, consistent with the condition given by

$$\det[\mathcal{H} - E\mathcal{S}] = 0, \tag{2.14}$$

where Eq. (2.14) is called the secular equation, and is an equation of degree n, whose solution gives all n eigenvalues of $E_i(\vec{k})$ $(i = 1, \cdots n)$ for a given $\vec{k}$.

[†]Since C_{ij} is generally a complex variable with two degrees of freedom, a real and a complex part, both C_{ij} and C_{ij}^* can be varied independently.

Using the expression for $E_i(\vec{k})$ in Eqs. (2.7) and (2.11), the coefficients $\mathbf{C}_i$ as a function of $\vec{k}$ are determined. In order to obtain the energy dispersion relations (or energy bands) $E_i(\vec{k})$, we solve the secular equation Eq. (2.14), for a number of high symmetry $\vec{k}$ points.

2.1.2 Procedure for obtaining the energy dispersion

In the tight binding method, the one-electron energy eigenvalues $E_i(\vec{k})$ are obtained by solving the secular equation Eq. (2.14). The eigenvalues $E_i(\vec{k})$ are a periodic function in the reciprocal lattice, which can be described within the first Brillouin zone. In a two or three dimensional solid, it is difficult to show the energy dispersion relations over the whole range of $\vec{k}$ values, and thus we plot $E_i(\vec{k})$ along the high symmetry directions in the Brillouin zone. The actual procedure of the tight binding calculation is as follows:

1. Specify the unit cell and the unit vectors, $\vec{a}_i$. Specify the coordinates of the atoms in the unit cell and select n atomic orbitals which are considered in the calculation.

2. Specify the Brillouin zone and the reciprocal lattice vectors, $\vec{b}_i$. Select the high symmetry directions in the Brillouin zone, and $\vec{k}$ points along the high symmetry axes.

3. For the selected $\vec{k}$ points, calculate the transfer and the overlap matrix element, $\mathcal{H}_{ij}$ and $\mathcal{S}_{ij}$.*

4. For the selected $\vec{k}$ points, solve the secular equation, Eq. (2.14) and obtain the eigenvalues $E_i(\vec{k})$ $(i = 1, \cdots, n)$ and the coefficients $C_{ij}(\vec{k})$.

Tight-binding calculations are not self-consistent calculations in which the occupation of an electron in an energy band would be determined self-consistently. That is, for given electron occupation, the potential of the Hamiltonian is calculated, from which the updated electron occupation is determined using, for example, Mulliken's gross population analysis [47]. When the input and the output of occupation of the electron are equal to each other within the desired accuracy, the eigenvalues are said to have been obtained self-consistently.

*When only the transfer matrix is calculated and the overlap matrix is taken as the unit matrix, then the Slater–Koster extrapolation scheme results.

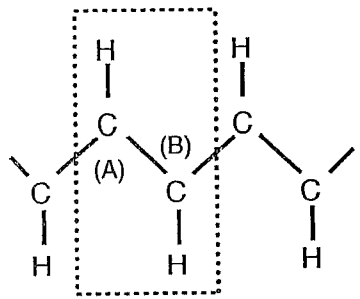


Fig. 2.1: The unit cell of *trans*-polyacetylene bounded by a box defined by the dotted lines, and showing two inequivalent carbon atoms, A and B, in the unit cell.

In applying these calculational approaches to real systems, the symmetry of the problem is considered in detail on the basis of a tight-binding approach and the transfer and the overlap matrix elements are often treated as parameters selected to reproduce the band structure of the solid obtained either experimentally or from first principles calculations. Both extrapolation methods such as $\vec{k}\cdot\vec{p}$ perturbation theory or interpolation methods using the Slater–Koster approach are commonly employed for carbon-related systems such as a 2D graphene sheet or 3D graphite [9].

2.2 Electronic Structure of Polyacetylene

A simple example of π-energy bands for a one-dimensional carbon chain is polyacetylene (see Sect. 1.2.2). In Fig. 2.1 we show, within the box defined by the dotted lines, the unit cell for *trans*-polyacetylene $(CH)_x$ which contains two inequivalent carbon atoms, A and B, in the unit cell. As discussed in Sect. 1.2.2, there is one π-electron per carbon atom, thus giving rise to two π-energy bands called bonding and anti-bonding π-bands in the first Brillouin zone.

The lattice unit vector and the reciprocal lattice vector of this one-dimensional molecule are given by $\vec{a}_1 = (a,0,0)$ and $\vec{b}_1 = (a/2\pi,0,0)$, respectively. The Brillouin zone is the line segment $-a/\pi < k < a/\pi$. The Bloch orbitals consisting of A and B atoms are given by

$$\Phi_j(r) = \frac{1}{\sqrt{N}}\sum_{R_\alpha} e^{ikR_\alpha}\varphi_j(r-R_\alpha), \quad (\alpha = \mathrm{A},\mathrm{B}) \tag{2.15}$$

where the summation is taken over the atom site coordinate R_α for the A or B

carbon atoms in the solid.

The (2×2) matrix Hamiltonian, $\mathcal{H}_{\alpha\beta}$, $(\alpha,\beta = A,B)$ is obtained by substituting Eq. (2.15) into Eq. (2.9). When $\alpha=\beta=$ A,

$$\begin{aligned}\mathcal{H}_{AA}(r) &= \frac{1}{N}\sum_{R,R'} e^{ik(R-R')}\langle\varphi_A(r-R')|\mathcal{H}|\varphi_A(r-R)\rangle \\ &= \frac{1}{N}\sum_{R=R'}\epsilon_{2p} + \frac{1}{N}\sum_{R=R'\pm a} e^{\pm ika}\langle\varphi_A(r-R')|\mathcal{H}|\varphi_A(r-R)\rangle \\ &\quad +(\text{terms equal to or more distant than } R=R'\pm 2a) \\ &= \epsilon_{2p} + (\text{terms equal to or more distant than } R=R'\pm a).\end{aligned} \tag{2.16}$$

In Eq. (2.16) the maximum contribution to the matrix element $\mathcal{H}_{AA}$ comes from $R=R'$, and this gives the orbital energy of the $2p$ level, ϵ_{2p}.* The next order contribution to $\mathcal{H}_{AA}$ comes from terms in $R=R'\pm a$, which will be neglected for simplicity. Similarly, $\mathcal{H}_{BB}$ also gives ϵ_{2p} for the same order of approximation.

Next let us consider the matrix element $\mathcal{H}_{AB}(r)$. The largest contribution to $\mathcal{H}_{AB}(r)$ arises when atoms A and B are nearest neighbors. Thus, in the summation over R', we only consider the cases $R'=R\pm a/2$ and neglect more distant terms to obtain

$$\begin{aligned}\mathcal{H}_{AB}(r) &= \frac{1}{N}\sum_{R}\Big\{e^{ika/2}\langle\varphi_A(r-R)|\mathcal{H}|\varphi_B(r-R-a/2)\rangle \\ &\qquad + e^{-ika/2}\langle\varphi_A(r-R)|\mathcal{H}|\varphi_B(r-R+a/2)\rangle\Big\} \\ &= 2t\cos(ka/2)\end{aligned} \tag{2.17}$$

where the transfer integral t is the integral appearing in Eq. (2.17) and denoted by†.

$$t = \langle\varphi_A(r-R)|\mathcal{H}|\varphi_B(r-R\pm a/2)\rangle. \tag{2.18}$$

It is stressed that t has a negative value. The matrix element $\mathcal{H}_{BA}(r)$ is obtained from $\mathcal{H}_{AB}(r)$ through the Hermitian conjugation relation $\mathcal{H}_{BA}=\mathcal{H}^*_{AB}$, but since $\mathcal{H}_{AB}$ is real, we obtain $\mathcal{H}_{BA}=\mathcal{H}_{AB}$.

*Note that ϵ_{2p} is not simply the atomic energy value for the free atom, because the Hamiltonian contains a crystal potential.

†Here we have assumed that all the π bonding orbitals are equal (1.5Å bonds). In the real $(CH)_x$ compound, bond alternation occurs, in which the bond energy alternates between 1.7Å and 1.3Å bonds, and the two atomic integrations in Eq. (2.17) are not equal. Although the distortion of the lattice lowers the energy, the electronic energy always decreases more than the lattice energy in a one-dimensional material, and thus the lattice becomes deformed by a process called the Peierls instability. See details in Sect. 11.3.1

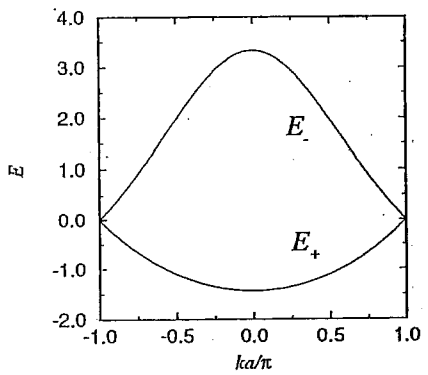


Fig. 2.2: The energy dispersion relation $E_{\pm}(\vec{k})$ for polyacetylene $[(CH)_x]$, given by Eq. (2.21) with values for the parameters $t = -1$ and $s = 0.2$. Curves $E_+(k)$ and $E_-(k)$ are called bonding π and antibonding π^* energy bands, respectively, and the plot is given in units of $|t|$.

The overlap matrix $\mathcal{S}_{ij}$ can be calculated by a similar method as used for $\mathcal{H}_{ij}$, except that the intra-atomic integral yields the energy for the crystal Hamiltonian $\mathcal{H}_{ij}$, but the overlap matrix rather yields unity for the case of $\mathcal{S}_{ij}$, if we assume that the atomic wavefunction is normalized, $\mathcal{S}_{AA} = \mathcal{S}_{BB} = 1$ and $\mathcal{S}_{AB} = \mathcal{S}_{BA} = 2s\cos(ka/2)$, where s is the overlap integral between the nearest A and B atoms,

$$s = \langle \varphi_A(r - R) | \varphi_B(r - R \pm a/2) \rangle. \tag{2.19}$$

The secular equation for the $2p_z$ orbital of $[(CH)_x]$ is given by

$$\begin{vmatrix} \epsilon_{2p} - E & 2(t - sE)\cos(ka/2) \\ 2(t - sE)\cos(ka/2) & \epsilon_{2p} - E \end{vmatrix} \tag{2.20}$$

$$= (\epsilon_{2p} - E)^2 - 4(t - sE)^2\cos^2(ka/2)$$
$$= 0,$$

yielding the eigenvalues of the energy dispersion relations of Eq. (2.20) given by

$$E_{\pm}(\vec{k}) = \frac{\epsilon_{2p} \pm 2t\cos(ka/2)}{1 \pm 2s\cos(ka/2)}, \quad (-\frac{\pi}{a} < k < \frac{\pi}{a}) \tag{2.21}$$

in which the + sign defines one branch and the − sign defines the other branch, as shown in Fig. 2.2, where we use values for the parameters, $\epsilon_{2p} = 0$, $t = -1$, and $s = 0.2$. The levels E_+ and E_- are degenerate at $ka = \pm\pi$.

$E_+(k)$ and $E_-(k)$ are called bonding π and antibonding π^* energy bands, respectively. Since there are two π electrons per unit cell, each with a different

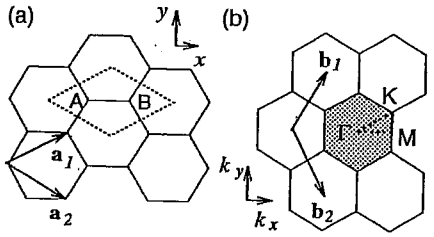


Fig. 2.3: (a) The unit cell and (b) Brillouin zone of two-dimensional graphite are shown as the dotted rhombus and the shaded hexagon, respectively. $\vec{a}_i$, and $\vec{b}_i$, $(i = 1, 2)$ are unit vectors and reciprocal lattice vectors, respectively. Energy dispersion relations are obtained along the perimeter of the dotted triangle connecting the high symmetry points, Γ, K and M.

spin orientation, both electrons occupy the bonding π energy band, which makes the total energy lower than ϵ_{2p}.

2.3 Two-Dimensional Graphite

Graphite is a three-dimensional (3D) layered hexagonal lattice of carbon atoms. A single layer of graphite, forms a two-dimensional (2D) material, called 2D graphite or a graphene layer. Even in 3D graphite, the interaction between two adjacent layers is small compared with intra-layer interactions, since the layer-layer separation of 3.35Å is much larger than nearest-neighbor distance between two carbon atoms, a_{C-C}=1.42Å. Thus the electronic structure of 2D graphite is a first approximation of that for 3D graphite.

In Fig. 2.3 we show (a) the unit cell and (b) the Brillouin zone of two-dimensional graphite as a dotted rhombus and shaded hexagon, respectively, where $\vec{a}_1$ and $\vec{a}_2$ are unit vectors in real space, and $\vec{b}_1$ and $\vec{b}_2$ are reciprocal lattice vectors. In the x, y coordinates shown in the Fig. 2.3, the real space unit vectors $\vec{a}_1$ and $\vec{a}_2$ of the hexagonal lattice are expressed as

$$\vec{a}_1 = \left(\frac{\sqrt{3}}{2}a, \frac{a}{2}\right), \quad \vec{a}_2 = \left(\frac{\sqrt{3}}{2}a, -\frac{a}{2}\right), \tag{2.22}$$

where $a = |\vec{a}_1| = |\vec{a}_2| = 1.42 \times \sqrt{3} = 2.46$Å is the lattice constant of two-dimensional graphite. Correspondingly the unit vectors $\vec{b}_1$ and $\vec{b}_2$ of the recip-

rocal lattice are given by:

$$\vec{b}_1 = \left(\frac{2\pi}{\sqrt{3}a}, \frac{2\pi}{a}\right), \quad \vec{b}_2 = \left(\frac{2\pi}{\sqrt{3}a}, -\frac{2\pi}{a}\right) \tag{2.23}$$

corresponding to a lattice constant of $4\pi/\sqrt{3}a$ in reciprocal space. The direction of the unit vectors $\vec{b}_1$ and $\vec{b}_2$ of the reciprocal hexagonal lattice are rotated by 90° from the unit vectors $\vec{a}_1$ and $\vec{a}_2$ of the hexagonal lattice in real space, as shown in Fig. 2.3. By selecting the first Brillouin zone as the shaded hexagon shown in Fig. 2.3(b), the highest symmetry is obtained for the Brillouin zone of 2D graphite. Here we define the three high symmetry points, Γ, K and M as the center, the corner, and the center of the edge, respectively. The energy dispersion relations are calculated for the triangle ΓMK shown by the dotted lines in Fig. 2.3(b).

As discussed in Sect. 2.3.2, three σ bonds for 2D graphite hybridize in a sp^2 configuration, while, and the other $2p_z$ orbital, which is perpendicular to the graphene plane, makes π covalent bonds. In Sect. 2.3.1 we consider only π energy bands for 2D graphite, because we know that the π energy bands are covalent and are the most important for determining the solid state properties of graphite.

2.3.1 π Bands of Two-Dimensional Graphite

Two Bloch functions, constructed from atomic orbitals for the two inequivalent carbon atoms at A and B in Fig. 2.3, provide the basis functions for 2D graphite. When we consider only nearest-neighbor interactions, then there is only an integration over a single atom in $\mathcal{H}_{AA}$ and $\mathcal{H}_{BB}$, as is shown in Eq. (2.16), and thus $\mathcal{H}_{AA} = \mathcal{H}_{BB} = \epsilon_{2p}$. For the off-diagonal matrix element $\mathcal{H}_{AB}$, we must consider the three nearest-neighbor B atoms relative to an A atom, which are denoted by the vectors $\vec{R}_1, \vec{R}_2$, and $\vec{R}_3$. We then consider the contribution to Eq. (2.17) from $\vec{R}_1, \vec{R}_2$, and $\vec{R}_3$ as follows:

$$\begin{aligned} \mathcal{H}_{AB} &= t(e^{i\vec{k}\cdot\vec{R}_1} + e^{i\vec{k}\cdot\vec{R}_2} + e^{i\vec{k}\cdot\vec{R}_3}) \\ &= tf(k) \end{aligned} \tag{2.24}$$

where t is given by Eq. (2.18)* and $f(k)$ is a function of the sum of the phase factors of $e^{i\vec{k}\cdot\vec{R}_j}$ $(j = 1, \cdots, 3)$. Using the x, y coordinates of Fig. 2.3(a), $f(k)$ is given by:

$$f(k) = e^{ik_x a/\sqrt{3}} + 2e^{-ik_x a/2\sqrt{3}} \cos\left(\frac{k_y a}{2}\right). \tag{2.25}$$

Since $f(k)$ is a complex function, and the Hamiltonian forms a Hermitian matrix, we write $\mathcal{H}_{BA} = \mathcal{H}^*_{AB}$ in which * denotes the complex conjugate. Using Eq. (2.25), the overlap integral matrix is given by $\mathcal{S}_{AA} = \mathcal{S}_{BB} = 1$, and $\mathcal{S}_{AB} = sf(k) = \mathcal{S}^*_{BA}$. Here s has the same definition as in Eq. (2.19), so that the explicit forms for $\mathcal{H}$ and $\mathcal{S}$ can be written as:

$$\mathcal{H} = \begin{pmatrix} \epsilon_{2p} & tf(k) \\ tf(k)^* & \epsilon_{2p} \end{pmatrix}, \quad \mathcal{S} = \begin{pmatrix} 1 & sf(k) \\ sf(k)^* & 1 \end{pmatrix}. \tag{2.26}$$

Solving the secular equation $\det(\mathcal{H} - E\mathcal{S}) = 0$ and using $\mathcal{H}$ and $\mathcal{S}$ as given in Eq. (2.26), the eigenvalues $E(\vec{k})$ are obtained as a function $w(\vec{k})$, k_x and k_y:

$$E_{g2D}(\vec{k}) = \frac{\epsilon_{2p} \pm tw(\vec{k})}{1 \pm sw(\vec{k})}, \tag{2.27}$$

where the + signs in the numerator and denominator go together giving the bonding π energy band, and likewise for the − signs, which give the anti-bonding π^* band, while the function $w(\vec{k})$ is given by:

$$w(\vec{k}) = \sqrt{|f(\vec{k})|^2} = \sqrt{1 + 4\cos\frac{\sqrt{3}k_x a}{2}\cos\frac{k_y a}{2} + 4\cos^2\frac{k_y a}{2}}. \tag{2.28}$$

In Fig. 2.4, the energy dispersion relations of two-dimensional graphite are shown throughout the Brillouin zone and the inset shows the energy dispersion relations along the high symmetry axes along the perimeter of the triangle shown in Fig. 2.3(b). Here we use the parameters $\epsilon_{2p} = 0$, $t = -3.033$eV, and $s = 0.129$ in order to reproduce the first principles calculation of the graphite energy bands [9, 48]. The upper half of the energy dispersion curves describes the π^*-energy anti-bonding band, and the lower half is the π-energy bonding band. The upper π^* band and the lower π band are degenerate at the K points through which

*We often use the symbol γ_0 for the nearest neighbor transfer integral. γ_0 is defined by a positive value.

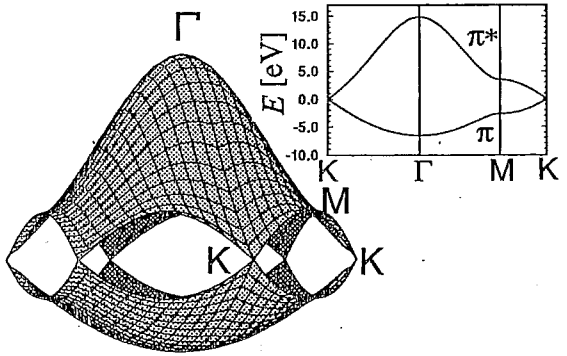


Fig. 2.4: The energy dispersion relations for 2D graphite are shown throughout the whole region of the Brillouin zone. The inset shows the energy dispersion along the high symmetry directions of the triangle ΓMK shown in Fig. 2.3(b) (see text).

the Fermi energy passes. Since there are two π electrons per unit cell, these two π electrons fully occupy the lower π band. Since a detailed calculation of the density of states shows that the density of states at the Fermi level is zero, two-dimensional graphite is a zero-gap semiconductor. The existence of a zero gap at the K points comes from the symmetry requirement that the two carbon sites A and B in the hexagonal lattice are equivalent to each other.† The existence of a zero gap at the K points gives rise to quantum effects in the electronic structure of carbon nanotubes, as shown in Chapter 3.

When the overlap integral s becomes zero, the π and π^* bands become symmetrical around $E = \epsilon_{2p}$ which can be understood from Eq. (2.27). The energy dispersion relations in the case of $s = 0$ (i.e., in the Slater–Koster scheme) are commonly used as a simple approximation for the electronic structure of a graphene layer:

$$E_{g2D}(k_x, k_y) = \pm t \left\{ 1 + 4\cos\left(\frac{\sqrt{3}k_x a}{2}\right)\cos\left(\frac{k_y a}{2}\right) + 4\cos^2\left(\frac{k_y a}{2}\right) \right\}^{1/2} . \tag{2.29}$$

†If the A and B sites had different atoms such as B and N, the site energy ϵ_{2p} would be different for B and N, and therefore the calculated energy dispersion would show an energy gap between the π and π^* bands.

In this case, the energies have the values of $\pm 3t$, $\pm t$ and 0, respectively, at the high symmetry points, Γ, M and K in the Brillouin zone. Thus the band width gives $|6t|$, which is consistent with the three connected π bonds. The simple approximation given by Eq. (2.29) is used in Sect. 4.1.2 to obtain a simple approximation for the electronic dispersion relations for carbon nanotubes.

2.3.2 σ Bands of Two-Dimensional Graphite

Finally let us consider the σ bands of two-dimensional graphite. There are three atomic orbitals of sp^2 covalent bonding per carbon atom, $2s$, $2p_x$ and $2p_y$. We thus have six Bloch orbitals in the 2 atom unit cell, yielding six σ bands. We will calculate these six σ bands using a 6×6 Hamiltonian and overlap matrix, and we will then solve the secular equation* for each $\vec{k}$ point. For the eigenvalues thus obtained, three of the six σ bands are bonding σ bands which appear below the Fermi energy, and the other three σ bands are antibonding σ^* bands above the Fermi energy.

The calculation of the Hamiltonian and overlap matrix is performed analytically, using a small number of parameters. Hereafter we arrange the matrix elements in accordance with their atomic identity for the free atom: $2s^A$, $2p_x^A$, $2p_y^A$, $2s^B$, $2p_x^B$, $2p_y^B$. Then the matrix elements coupling the same atoms (for example A and A) can be expressed by a 3×3 small matrix which is a sub-block of the 6×6 matrix. Within the nearest neighbor site approximation given by Eq. (2.16), the small Hamiltonian and overlap matrices are diagonal matrices as follows,

$$\mathcal{H}_{AA} = \begin{pmatrix} \epsilon_{2s} & 0 & 0 \\ 0 & \epsilon_{2p} & 0 \\ 0 & 0 & \epsilon_{2p} \end{pmatrix}, \quad \mathcal{S}_{AA} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{2.30}$$

where ϵ_{2p} is defined by Eq. (2.16) and ϵ_{2s} is the orbital energy of the $2s$ levels.

The matrix element for the Bloch orbitals between the A and B atoms can be obtained by taking the components of $2p_x$ and $2p_y$ in the directions parallel or perpendicular to the σ bond. In Fig. 2.5, we show how to rotate the $2p_x$ atomic orbital and how to obtain the σ and π components for the rightmost

*Since the planar geometry of graphite satisfies the even symmetry of the Hamiltonian $\mathcal{H}$ and of $2s$, $2p_x$ and $2p_y$ upon mirror reflection about the xy plane, and the odd symmetry of $2p_z$, the σ and π energy bands can be solved separately, because matrix elements of different symmetry types do not couple in the Hamiltonian.

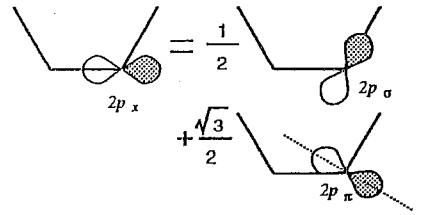


Fig. 2.5: The rotation of $2p_x$. The figure shows how to project the σ and π components along the indicated bond starting with the $2p_x$ orbital. This method is valid only for p orbitals.

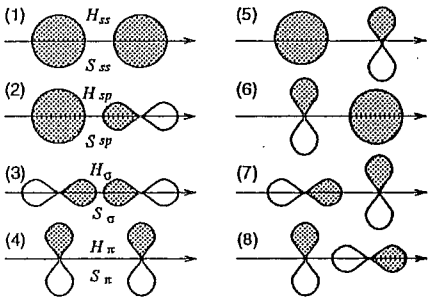


Fig. 2.6: The band parameters for σ bands. The four cases from (1) to (4) correspond to matrix elements having non-vanishing values and the remaining four cases from (5) to (8) correspond to vanishing matrix elements.

bond of this figure. In Fig. 2.5 the wavefunction of $|2p_x\rangle$ is decomposed into its σ and π components as follows:

$$|2p_x\rangle = \cos\frac{\pi}{3}|2p_\sigma\rangle + \sin\frac{\pi}{3}|2p_\pi\rangle. \tag{2.31}$$

This type of decomposition can be used to describe a bond in any general direction, which is discussed in Sect. 1.2. This procedure is also useful for fullerenes and carbon nanotubes, when we consider the curvature of their surfaces.

By rotating the $2p_x$ and $2p_y$ orbitals in the directions parallel and perpendicular to the desired bonds, the matrix elements appear in only 8 patterns as shown in Fig. 2.6, where shaded and not-shaded regions denote positive and negative amplitudes of the wavefunctions, respectively. The four cases from (1) to (4) in Fig. 2.6 correspond to non-vanishing matrix elements and the remaining four cases from (5) to (8) correspond to matrix elements which vanish because of symmetry. The corresponding parameters for both the Hamiltonian and the overlap matrix elements are shown in Fig. 2.6.

In Figs. 2.7(a) and (b) we show examples of the matrix elements of $\langle 2s^A|\mathcal{H}|2p_x^B\rangle$ and $\langle 2p_x^A|\mathcal{H}|2p_y^B\rangle$, respectively, obtained by the methods described above. In the case of Fig. 2.7 (a), there is only one non-vanishing contribution, and this

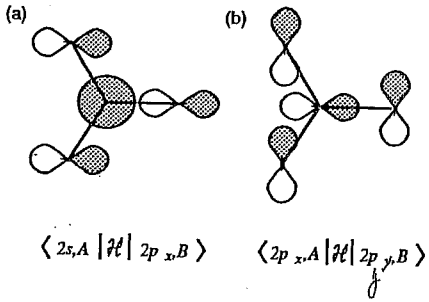


Fig. 2.7: Examples of the Hamiltonian matrix elements of σ orbitals, (a) $\langle 2s^A|\mathcal{H}|2p_x^B\rangle$ and (b) $\langle 2p_x^A|\mathcal{H}|2p_y^B\rangle$. By rotating the $2p$ orbitals, we get the matrix elements in Eq. (2.32) and Eq. (2.33), respectively.

comes from Fig. 2.6 (2). The other pertinent cases of Fig. 2.6 (5) or (6) give matrix elements that vanish by symmetry. Multiplying the phase factors for the three nearest neighbor B atoms with the matrix elements, we get the following result:

$$\langle 2s^A|\mathcal{H}|2p_x^B\rangle = \mathcal{H}_{sp}\left(-e^{ik_x a/\sqrt{3}} + e^{-ik_x a/2\sqrt{3}}\cos\frac{k_y a}{2}\right). \tag{2.32}$$

Similarly, in the case of Fig. 2.7 (b), the non-zero matrix elements correspond to the cases of Fig. 2.6 (3) and (4),

$$\begin{aligned}&\langle 2p_x^A|\mathcal{H}|2p_y^B\rangle \\ &= \tfrac{\sqrt{3}}{4}(\mathcal{H}_\sigma + \mathcal{H}_\pi)e^{-ik_x a/2\sqrt{3}}e^{ik_y a/2} - \tfrac{\sqrt{3}}{4}(\mathcal{H}_\sigma + \mathcal{H}_\pi)e^{-ik_x a/2\sqrt{3}}e^{-ik_y a/2} \\ &= \tfrac{\sqrt{3}i}{2}(\mathcal{H}_\sigma + \mathcal{H}_\pi)e^{-ik_x a/2\sqrt{3}}\sin\tfrac{k_y a}{2}.\end{aligned} \tag{2.33}$$

The resulting matrix element in Eq. (2.33) is a pure imaginary. However, the calculated results for the energy eigenvalues give real values.

When all the matrix elements of the 6×6 Hamiltonian and overlap matrices are calculated in a similar way, the 6×6 Hamiltonian matrix is obtained as a function of k_x and k_y. For given $\vec{k}$ points we then calculate the energy dispersion of the σ bands from the secular equation of Eq. (2.14). The results thus obtained for the calculated σ and π energy bands are shown in Fig. 2.8. Here we have used the parameters listed in Table 2.1, yielding a fit of the functional form of the energy bands imposed by symmetry to the energy values obtained for the first principles band calculations at the high symmetry points [48].

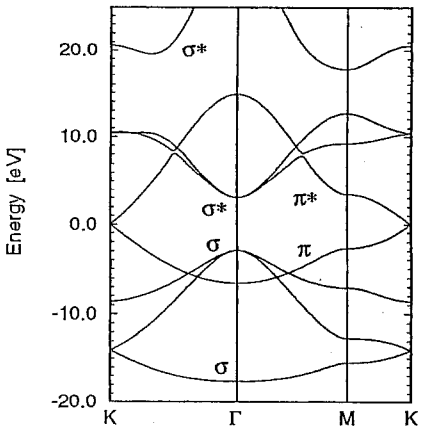


Fig. 2.8: The energy dispersion relations for σ and π bands of two-dimensional graphite. Here we used the parameters listed in Table 2.1.

Table 2.1: Values for the coupling parameters for carbon atoms in the Hamiltonian for π and σ bands in 2D graphite.

$\mathcal{H}$	value (eV)	$\mathcal{S}$	value
$\mathcal{H}_{ss}$	−6.769	$\mathcal{S}_{ss}$	0.212
$\mathcal{H}_{sp}$	−5.580	$\mathcal{S}_{sp}$	0.102
$\mathcal{H}_{\sigma}$	−5.037	$\mathcal{S}_{\sigma}$	0.146
$\mathcal{H}_{\pi} \equiv t$	−3.033	$\mathcal{S}_{\pi} \equiv s$	0.129
$\epsilon_{2s}{}^{(a)}$	−8.868		

[a]The value for ϵ_{2s} is given relative to setting $\epsilon_{2p} = 0$.

In the absence of more detailed experimental or theoretical information, these parameters can be used as a first approximation in describing the matrix elements for most sp^2 carbon materials for which the carbon-carbon distance is close to that of graphite, 1.42Å. Further, if the parameters are only slightly changed, the formulation can be used to describe the sp^3 diamond system and sp carbyne materials.

As is shown in Fig. 2.8, the π and the two σ bands cross each other (i.e., have different symmetries), as do the π^* and the two σ^* bands. However, because of the different group theoretical symmetries between σ and π bands, no band separation occurs at the crossing points. The relative positions of these crossings are known to be important for: (1) photo-transitions from σ to π^* bands and

from π to σ^* bands, which satisfy the selection rule for electric dipole transitions, and (2) charge transfer from alkali metal ions to graphene sheets in graphite intercalation compounds.

Using the basic concepts of two-dimensional graphite presented in Chapter 2, we next discuss the structure and electronic properties of single-wall carbon nanotubes in Chapters 3 and 4, respectively.